Quiz App

Software Development Project Report

Project Type: Java Desktop Application
Technology: Java Swing, Object-Oriented Programming
Development Period: 10 Weeks

Final Project Report

1. Introduction

The Quiz App is a Java-based interactive desktop application developed as a project for the Software Development I course. The main purpose of this software is to provide an interactive environment where a user can answer multiple-choice questions and receive result information after completing the quiz.

The application uses Java Swing to create the graphical user interface and follows Object-Oriented Programming concepts to organize the different components of the system. The software provides a graphical quiz interface, question display, multiple-choice answer selection, keyboard/mouse interaction through GUI controls, next-question navigation, response storage and result processing.

The project was developed incrementally. Different features were implemented, tested and improved step by step throughout the development period.



2. Project Objectives

• To develop an interactive quiz application using Java.
• To create a simple and user-friendly graphical user interface.
• To display multiple-choice questions and answer options.
• To allow the user to select an answer using radio buttons.
• To move from one question to the next using a Next button.
• To store the user's selected responses.
• To process the stored responses and generate result information.
• To apply Object-Oriented Programming concepts.
• To use Java Swing components for GUI development.
• To use GitHub for source-code management and project development

3. Project Scope

The project focuses on developing a desktop-based quiz application using Java and Java Swing. The user can start the application and answer a sequence of multiple-choice questions through a graphical interface.

The system displays each question with four answer choices. The user selects an option and proceeds using the Next button. The selected responses are stored so that the application can process them after the quiz. The project is designed as a standalone Java application and does not require a database.

4. Development Methodology

The project was developed using the Incremental Development Model. In this model, the complete system is divided into smaller increments, and each increment adds a specific functionality to the software.

The project was planned and developed through several stages

Week	Development Work
Week 1	Project Planning and Requirement Analysis
Week 2	Software Design
Week 3	User Interface Development
Week 4	Aircraft Management System
Week 5	Runway Allocation System
Week 6	Integration and Improvement of Core Module
Week 7	Emergency Handling System
Week 8	Status Monitoring and Reporting
Week 9	Software Finalization and UI Improvement
Week 10	Testing and Final Release

5. Software and Hardware Requirements

5.1 Software Requirements

• Java Development Kit (JDK)
• Intellij Idea
• Java Swing
• GitHub
• Windows Operating System

5.2 Hardware Requirements

The application does not require high-end hardware. A normal computer or laptop capable of running Java and Intellij Idea is sufficient.

• Computer or Laptop
• Keyboard and Mouse
• Sufficient RAM
• Sufficient storage space

6. System Features

The major features of the Quiz App are described below.

6.1 Quiz Interface

The Quiz Interface is the main interaction area of the application. It displays a question and a set of multiple-choice answer options. The user can select one option and continue to the next question.

The Runway Management option opens runway-related functions such as adding, removing, viewing, and assigning runways.



6.2 Question Display

The application stores questions in a String array and displays them one by one. The sample application contains questions such as the capital of Bangladesh, a language used for web development, and the meaning of CPU.

6.3 Answer Options

Each question contains four answer choices. Radio buttons are used so that the user can select one option from the available answers.

6.4 User Response Handling

The selected option is stored in a user Responses array. This allows the application to remember the user's selections while moving between questions.

6.5 Next Button

The Next button controls question navigation. After the user selects an option, pressing Next moves the application to the next question.



6.6 Result Generation

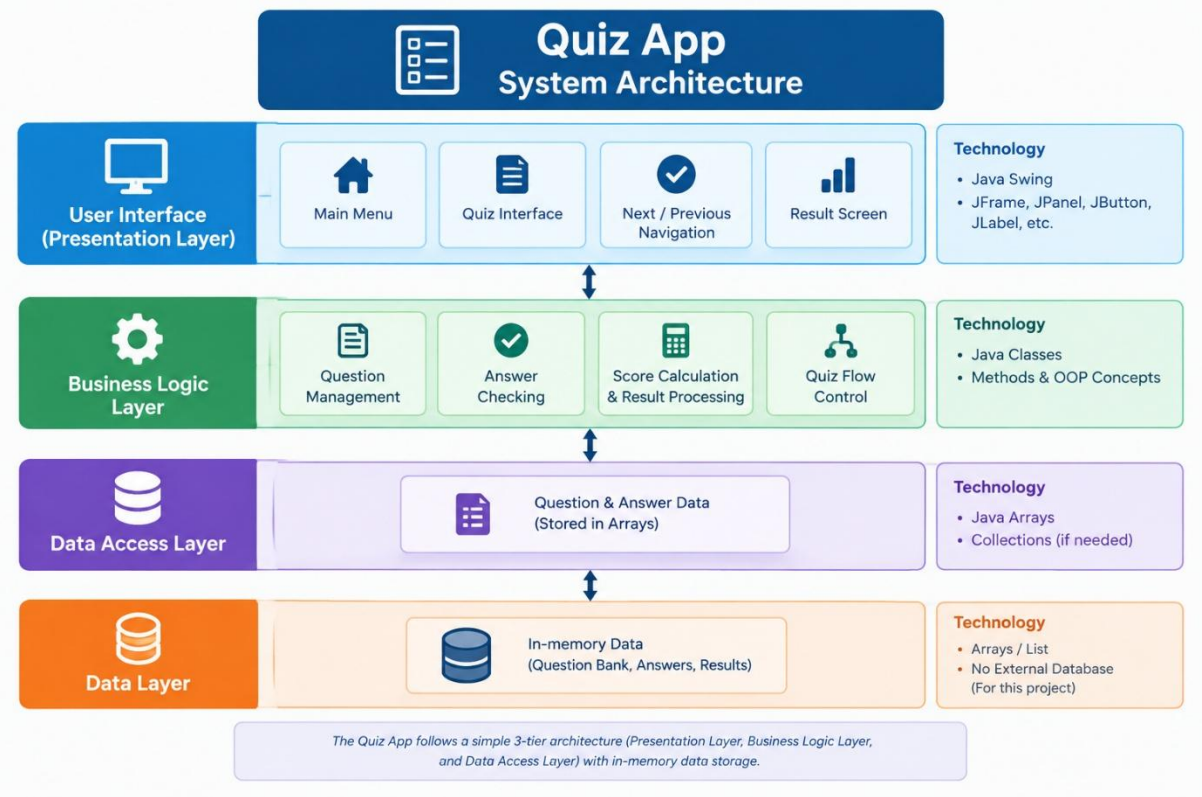
After the user completes the questions, the stored responses can be checked against the correct answers. The application then generates result information based on the user's responses

6.7 Multiple-Choice Interaction

The use of JRadioButton provides a simple and familiar interaction method for multiple-choice questions. The interface allows the user to focus on one question at a time

7. System Architecture

The system is organized into GUI components, question data and response-processing logic



8. Project Structure

The Java source files are organized into logical classes and modules. The project structure separates user-interface components from airport-management logic.

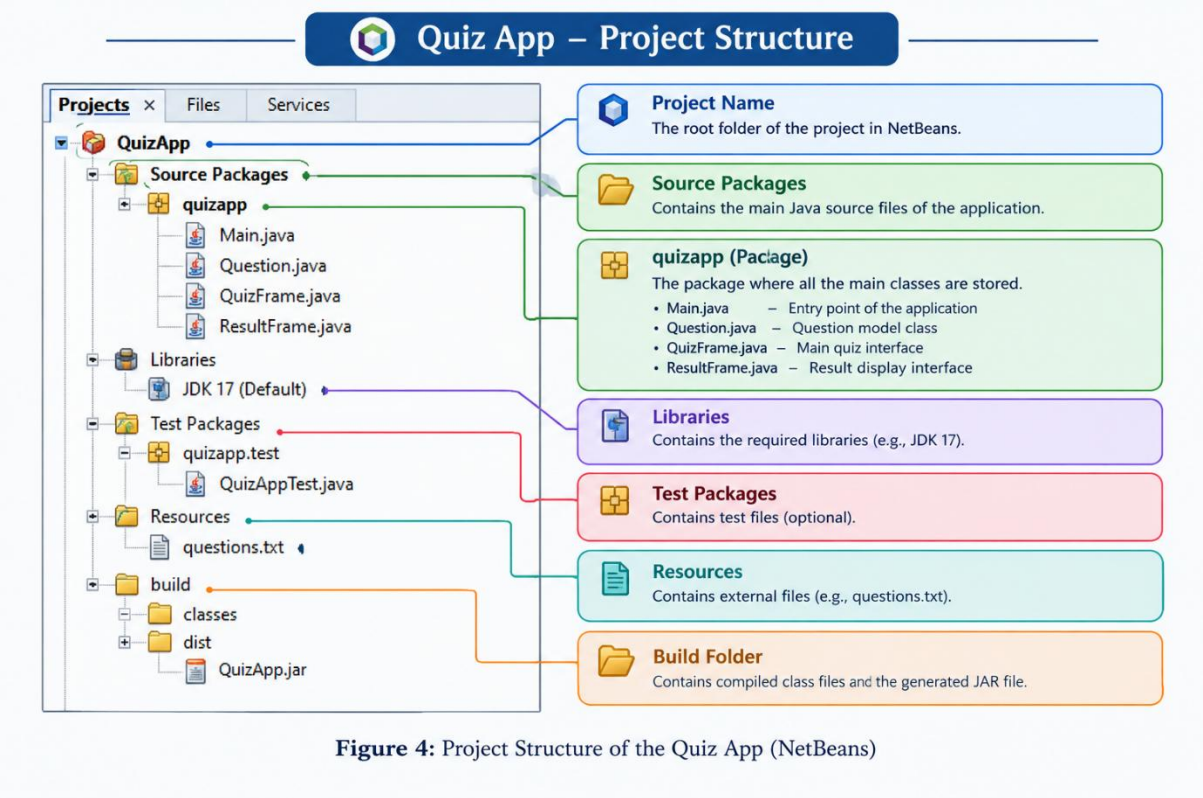


Figure 4: Project Structure of the Quiz App (NetBeans)

9. Object-Oriented Programming Concepts Used

The project applies several important Object-Oriented Programming concepts.

9.1 Class and Object

The application is organized through a Java class. An object of the GUI class is created to run the application.

9.2 Inheritance

The main GUI class inherits from JFrame. For example: public class Main extends JFrame. This allows the class to use JFrame functionality.

9.3 Encapsulation

Application data such as questions, answers and user responses are grouped within the class and are managed through the program's event-handling logic.

9.4 Modularity

Different responsibilities such as displaying questions, handling answer selection and navigating between questions are handled as separate logical parts of the application.

10. Data Structures Used

The application uses arrays to store question, answer and response information.

Questions are stored using a one-dimensional String array. The available answers are stored using a two-dimensional String array, where each question has four choices. User selections are stored using a userResponses array.

Example structure:

String[] questions = { "What is the capital of Bangladesh?", "Which language is used for web development?", "What does CPU stand for?" };
String[][] answers = { {"Dhaka", "Khulna", "Rajshahi", "Sylhet"}, ... };

11 Quiz Rules

• The quiz starts with the first question.
• The user selects one answer from the available options.
• The selected response is stored.
• The Next button moves the quiz to the following question.
• The quiz continues until the final question.
• After the final question, the stored responses are processed.
• The final result is generated from the responses.

12. System Workflow

The overall workflow of the Airport Runway Management Simulator can be represented as follows:

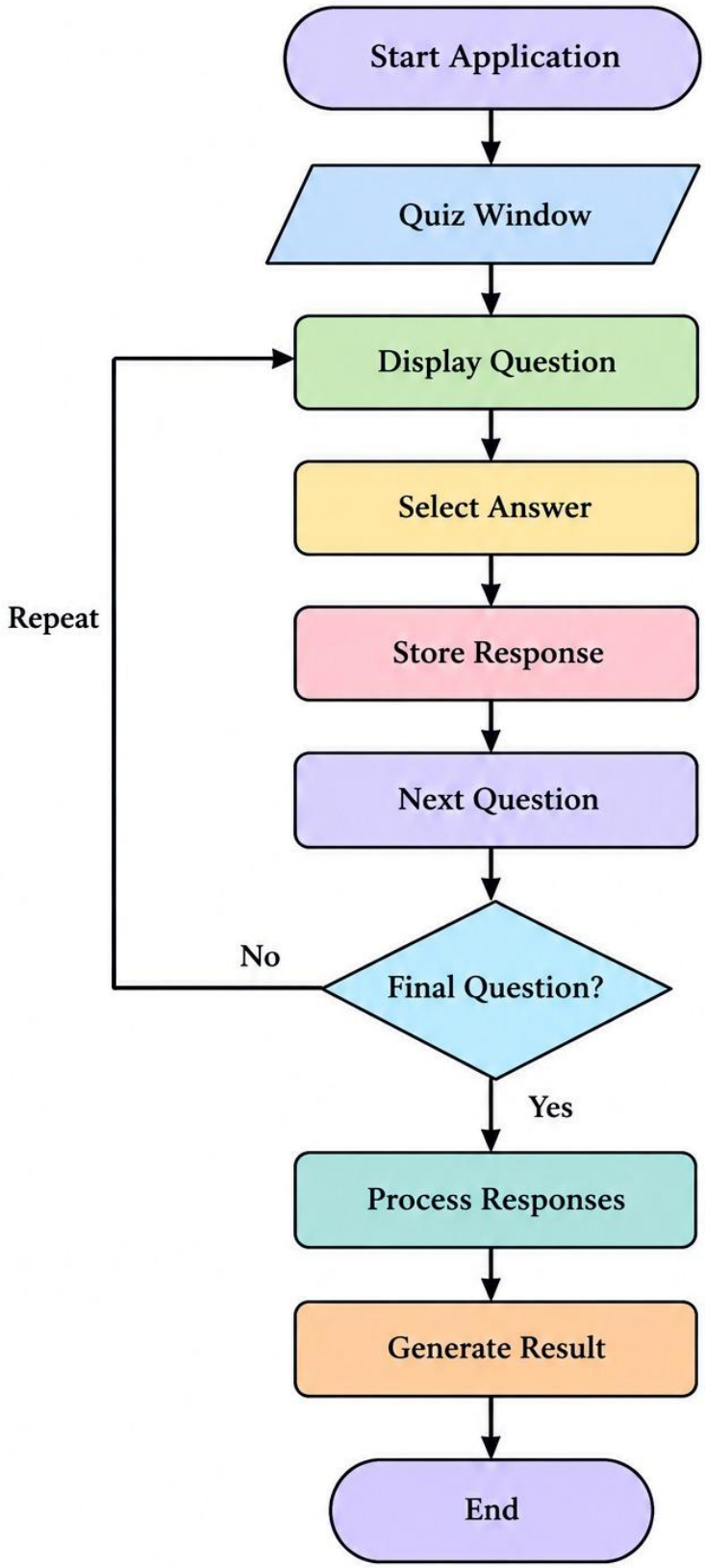


Figure 6: Quiz App Flow Diagram

13. Testing

Testing was performed to verify the functionality of the software. The main test cases are summarized below.

Test Case	Expected Result
Start Application	Quiz Windows open
Display Question	Question appears correctly
Select Option	Selected radio button is identified
Next Button	Next question appears
Response Storage	Selected response is retained
Final Question Quiz	Quiz proceeds to result processing
Result Processing	Result information is generated

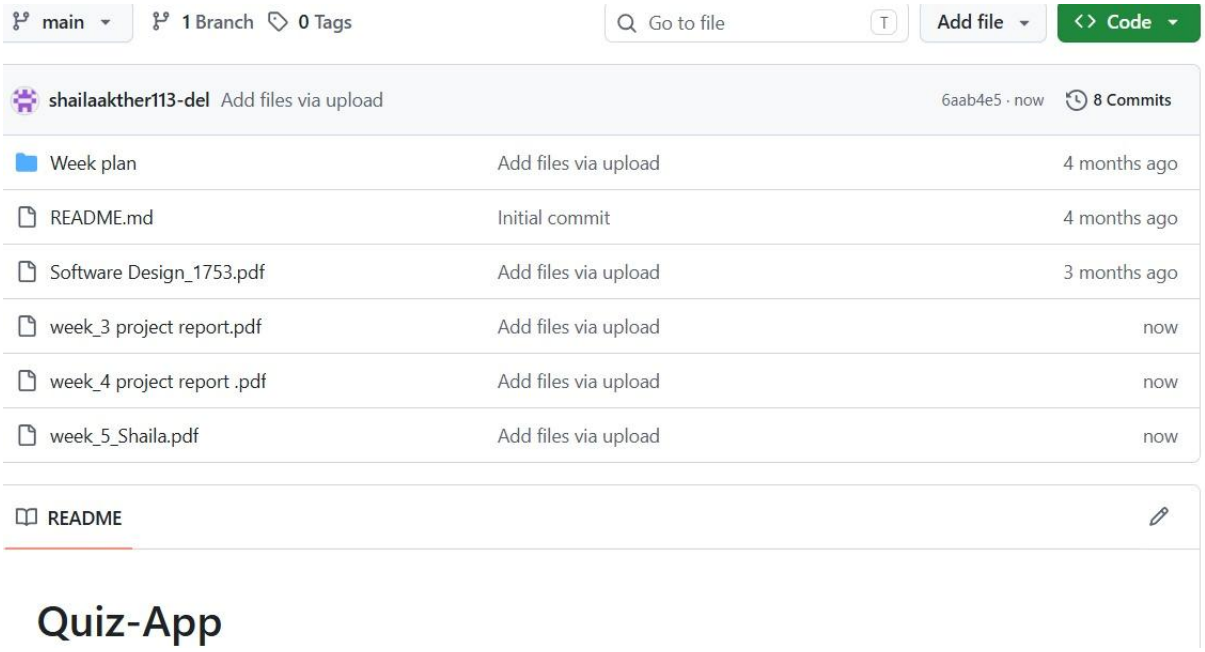
14. Error Handling and Validation

The application validates user interaction before moving through the quiz. The system checks whether an option has been selected and processes the response before continuing.

This validation logic helps prevent incomplete interaction and keeps the question sequence consistent

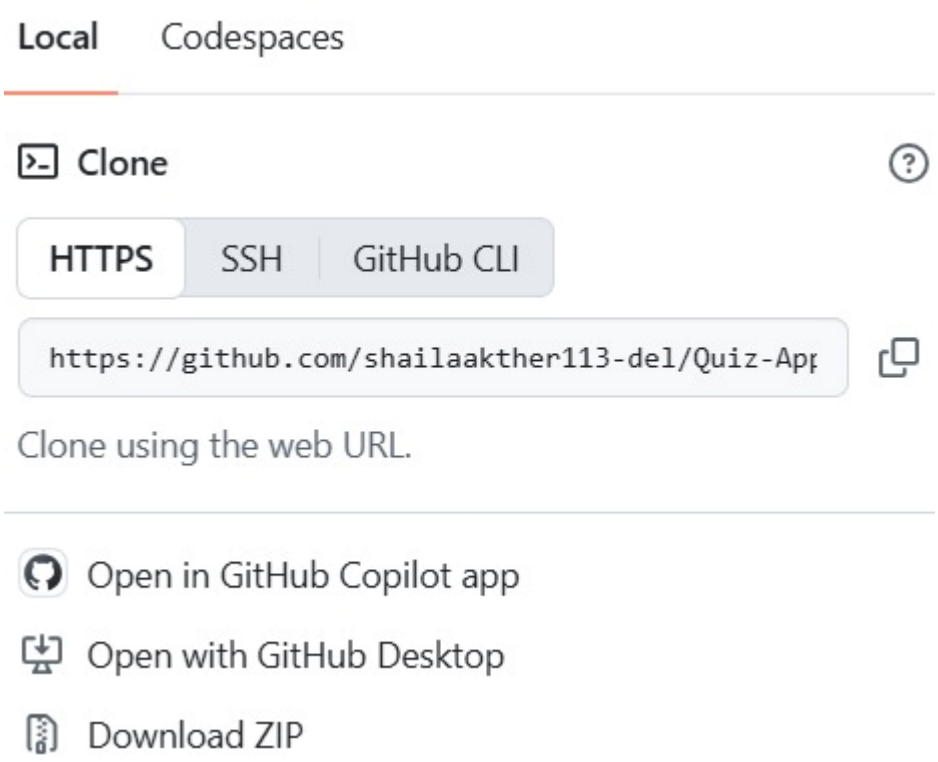
15. GitHub Version Control

GitHub can be used to store and manage the source code and project documentation. During development, different project files and modules can be uploaded to the repository according to the development schedule..



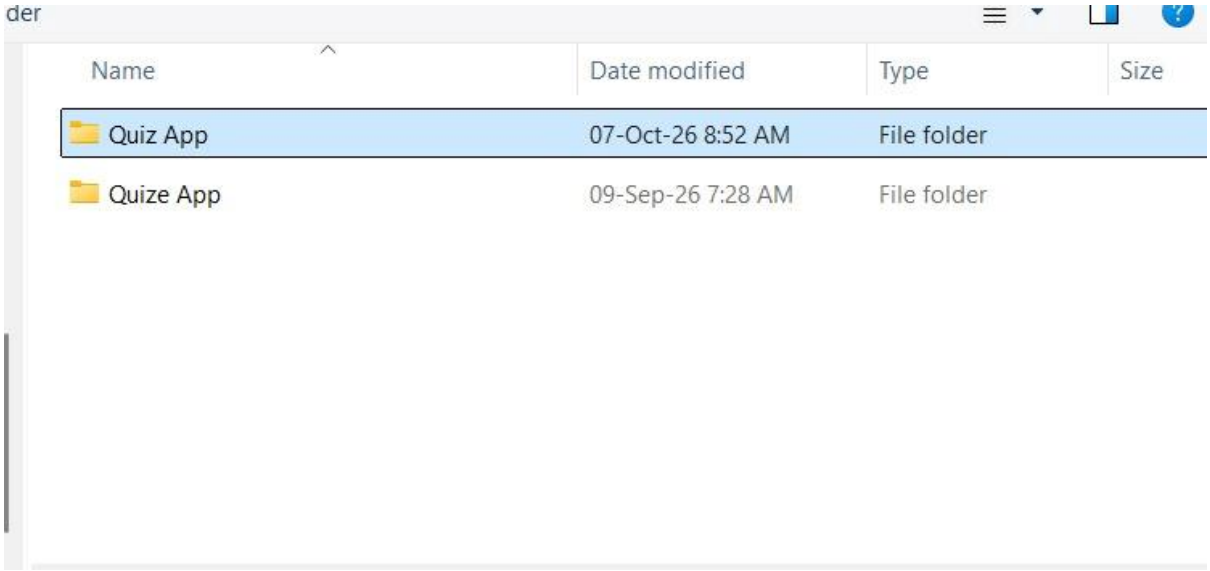
16. Downloading the Project from GitHub

Another user can download and use the Quiz App from the GitHub repository.
Step 1: Open GitHub Repository
Open the Quiz App repository on GitHub.
Step 2: Click Code
Click the Code button on the repository page.
Step 3: Download ZIP
Select Download ZIP and save the project to the computer.
Step 4: Extract the ZIP File
Extract the downloaded ZIP file.



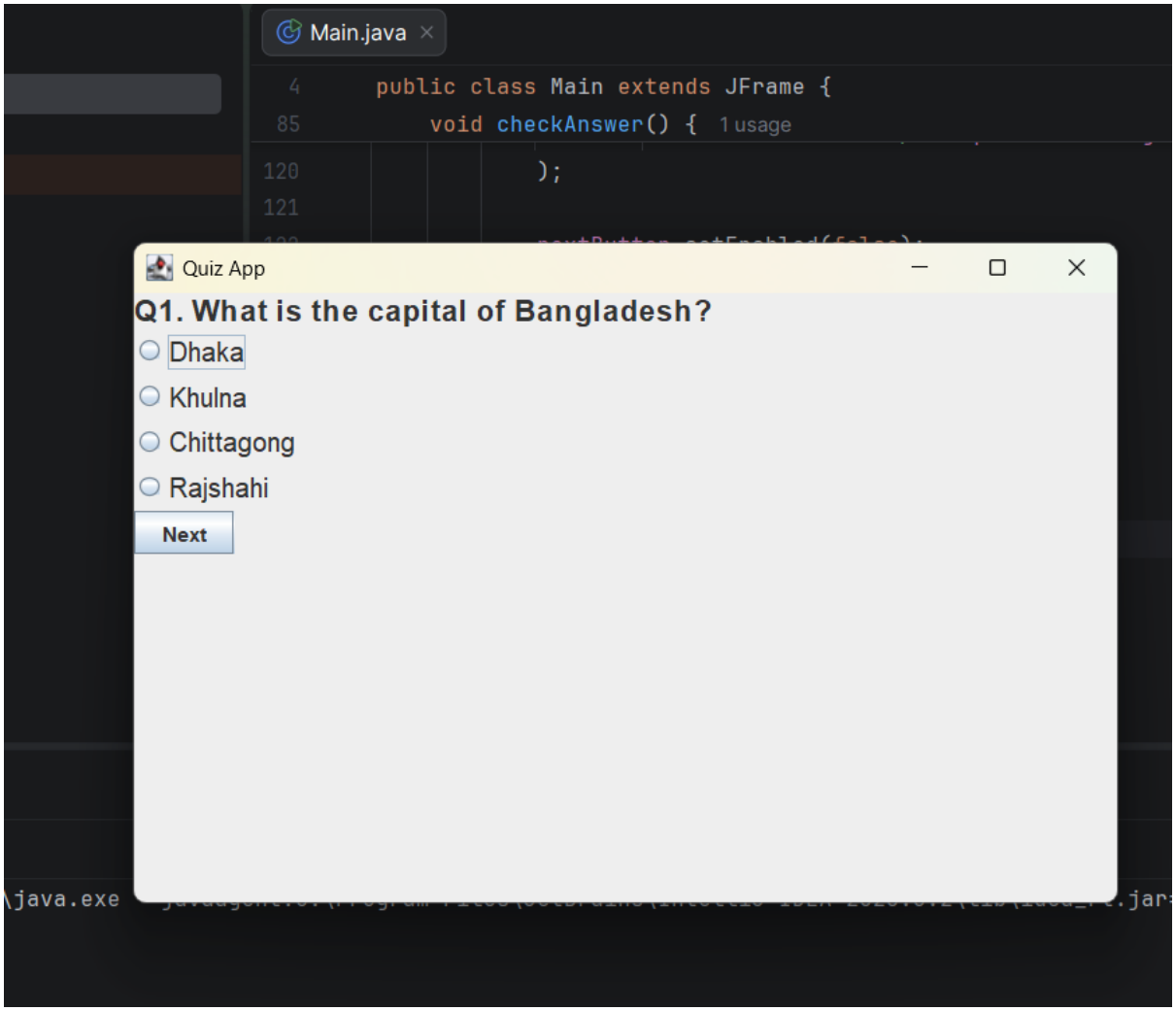
17. Opening the Project in Intellij Idea

- Open intellij idea.
- Select File → Open Project.
- Select the extracted Quiz App project folder.
- Click Open Project.
- Wait for the project to load.
- Check the Java source files.



18. Running the Project

- Run the project.
- The Quiz App window will appear.
- Read the displayed question.
- Select one of the available answers.
- Click Next to continue.
- Complete the remaining questions.
- View the generated result.



19. How Another User Can Use the Software

The complete process for another user is:
GitHub Repository

↓

Download ZIP

↓

Extract Project

↓

Open NetBeans

↓

Open Project

↓

Run Project

↓

Quiz Window

↓

Answer Questions

↓

View Result

Therefore, another user does not need to create the project from the beginning. The user can download the source code from GitHub, open it in NetBeans and run the application.

20. Challenges Faced

• Designing the graphical quiz interface.
• Displaying questions and answer options correctly.
• Connecting radio buttons with question data.
• Storing user responses correctly.
• Managing question navigation.
• Handling the final question and result processing.
• Connecting GUI components with Java event handling.
• Testing and debugging the complete application

These challenges provided practical experience in Java programming, GUI development and software development.

21. Result

The Quiz App was successfully developed as an interactive Java Swing application. The final application provides a graphical interface where users can answer multiple-choice questions.

The system displays questions and answer options, allows users to select responses through radio buttons, stores the selected responses and provides navigation between questions. The result-processing part uses the stored responses to generate result information.

The project also demonstrates the practical use of Java, Object-Oriented Programming, Java Swing, event handling, arrays, GUI components and GitHub version control.

.

22. Future Improvements

• Add a larger question bank.
• Add different quiz categories.
• Add random question selection.
• Add a timer for each quiz.
• Add difficulty levels.
• Add a score and performance summary.
• Add a restart quiz option.
• Add a database for storing questions.
• Add user login and profile features.
• Improve the graphical design and animations.
• Add sound effects.
• Add dark mode

23. Conclusion

The Quiz App project provided practical experience in the complete software development process. The project started with requirement analysis and design and was gradually developed through GUI development, question and answer implementation, response handling, navigation, result processing, testing and documentation.

The project demonstrates the practical application of Java, Object-Oriented Programming, Java Swing, event handling, arrays, GUI components and GitHub version control.

The final application provides an interactive environment where users can answer multiple-choice questions and receive result information after completing the quiz.

24. Project Demonstration

During the final project demonstration, the following features will be shown:
• Launching the application.
• Quiz App interface.
• Displaying a question.
• Selecting an answer using radio buttons.
• Moving to the next question.
• Storing user responses.
• Completing the quiz.
• Processing the final responses.
• Displaying the result.
• GitHub repository.
• Opening the project in Intellij idea.

25. Final Release

The final version of the Quiz App is prepared as Version 1.0.
The final project repository can contain

- Final Source Code
- Project Documentation
- Software Design Information
- Workflow Diagram
- User Manual
- Screenshots
- README file
- Final Project Files

The project can be downloaded from GitHub and opened in NetBeans for execution and further development.